



GyaanVault

SPPU M.Sc. Data Science — Free Study Material

Deep Learning

DS-562-MJ

Interview Preparation — Topic-wise Q&A

Semester II · Elective

Curated interview questions and answers for Deep Learning, organised by topic from the SPPU M.Sc. Data Science syllabus. Ideal for viva, placement and internship preparation.

Tip: read the answer, then close the PDF and try to explain each concept aloud in your own words.

Neural network basics

Q. What is a perceptron?

The simplest neural unit — it computes a weighted sum of inputs, adds a bias, and applies an activation function to produce an output.

Q. What is backpropagation?

An algorithm that computes gradients of the loss w.r.t. weights using the chain rule and updates weights via gradient descent.

Q. Name common activation functions.

Sigmoid, tanh, ReLU (and variants like Leaky ReLU), and softmax for multi-class output layers.

Q. What is the vanishing gradient problem?

In deep networks, gradients shrink through many layers (especially with sigmoid/tanh), slowing learning. ReLU and architectures like LSTM mitigate it.

CNN & RNN

Q. What is a CNN and where is it used?

A Convolutional Neural Network uses convolution and pooling layers to learn spatial features — ideal for image classification and computer vision.

Q. What are convolution, pooling and stride?

Convolution applies filters to detect features; pooling downsamples feature maps; stride is the step size of the filter.

Q. What is transfer learning?

Reusing a pre-trained model (e.g. VGG, ResNet) and fine-tuning it on a new, smaller dataset to save time and improve accuracy.

Q. What is an RNN / LSTM used for?

Recurrent networks model sequential data (text, time-series); LSTM/GRU add gates to remember long-term dependencies and avoid vanishing gradients.

